Qudit Stabilizer Simulation Reference

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***Abstract*—A self-contained reference for tableau-based simulation of Clifford circuits on qudits of prime or composite dimension d. Covers the stabilizer representation, gate conjugation rules, and a measurement protocol using Smith Normal Form reduction.**

***Index Terms*—Clifford Circuit, Smith Normal Form, Tableau Representation, Qudit**

I. Representation

A Pauli operator on n qudits of dimension d is defined as

$$P = \tau^{-\varphi} Z^{\mathbf{z}} X^{\mathbf{x}}, \quad \mathbf{z}, \mathbf{x} \in \mathbb{Z}_d^n, \quad \varphi \in \mathbb{Z}_D \tag{1}$$

where $\omega = e^{2\pi i/d}$, $\tau = e^{i\pi(d^2+1)/d}$, $\tau^2 = \omega$, and

$$D = \begin{cases} d & d \text{ odd} \\ 2d & d \text{ even.} \end{cases} \tag{2}$$

Two Paulis thus satisfy the commutation relation

$$P_1 P_2 = \tau^{2\langle 1,2\rangle} P_2 P_1, \tag{3}$$

where $\langle 1, 2\rangle$ is shorthand for the symplectic inner product of the Pauli vectors of P_1 and P_2:

$$\begin{aligned} \langle 1, 2\rangle &:= \langle \mathbf{z}_1 \oplus \mathbf{x}_1, \mathbf{z}_2 \oplus \mathbf{x}_2 \rangle \\ &= \mathbf{z}_1 \cdot \mathbf{x}_2 - \mathbf{x}_1 \cdot \mathbf{z}_2 \pmod d \end{aligned} \tag{4}$$

with $\oplus$ denoting the direct sum of vectors and $(\cdot)$ an inner product. Thus the Paulis commute iff $\langle 1, 2\rangle = 0$. Throughout, two identities will be used. First the *ordered product rule*:

$$\begin{aligned} &(\tau^{-\varphi_1} Z^{\mathbf{z}_1} X^{\mathbf{x}_1})(\tau^{-\varphi_2} Z^{\mathbf{z}_2} X^{\mathbf{x}_2}) \\ &= \tau^{-(\varphi_1+\varphi_2+2\mathbf{x}_1\cdot\mathbf{z}_2)} Z^{\mathbf{z}_1+\mathbf{z}_2} X^{\mathbf{x}_1+\mathbf{x}_2} \end{aligned} \tag{5}$$

and the *power rule*:

$$(\tau^{-\varphi} Z^{\mathbf{z}} X^{\mathbf{x}})^c = \tau^{-(c\varphi + c(c-1)\mathbf{z}\cdot\mathbf{x})} Z^{c\mathbf{z}} X^{c\mathbf{x}}. \tag{6}$$

The state is encoded in a column-major tableau with ZX ordering shown in Fig. 1. Each column is one stabilizer generator $S_j = \tau^{-\varphi_j} Z^{\mathbf{z}_j} X^{\mathbf{x}_j}$. Equivalently,

$$T \in \mathbb{Z}_D^{1\times\ell} \oplus \mathbb{Z}_d^{n\times\ell} \oplus \mathbb{Z}_d^{n\times\ell}, \quad \ell \le 2n, \tag{7}$$

Writing those blocks explicitly:

$$\begin{array}{c} \text{phase} \\ \\ \boldsymbol{Z} \\ \\ \\ \boldsymbol{X} \\ \\ \end{array} \left[\begin{array}{c} \varphi_1 \cdots \varphi_\ell \\ \hline z_{1,1} \cdots z_{1,\ell} \\ \vdots \qquad \vdots \\ z_{n,1} \cdots z_{n,\ell} \\ \hline x_{1,1} \cdots x_{1,\ell} \\ \vdots \qquad \vdots \\ x_{n,1} \cdots x_{n,\ell} \end{array}\right] \begin{array}{l} \in \mathbb{Z}_D^{1\times\ell} \\ \\ \left.\begin{array}{c} \\ \\ \\ \end{array}\right\} n \text{ rows} \in \mathbb{Z}_d^{n\times\ell} \\ \\ \left.\begin{array}{c} \\ \\ \\ \end{array}\right\} n \text{ rows} \in \mathbb{Z}_d^{n\times\ell} \end{array}$$

Fig. 1: Stabilizer tableau layout.

For the initial state $|0\rangle^{\otimes n}$ with $\ell = n$, we set the tableau to have the following form:

$$T: \quad \boldsymbol{\varphi} = \mathbf{0}, \quad \boldsymbol{Z} = \mathbf{1}_n, \quad \boldsymbol{X} = \mathbf{0}_n \tag{8}$$

For composite d, the number of generators ℓ can grow beyond n (up to $2n$) after measurements. For prime d, $\ell = n$ always.

A. Generator Arithmetic

Two column-level operations are needed. To *multiply* generator k by c copies of generator p (implementing $S_k \leftarrow S_p^c S_k$), apply

$$\begin{aligned} \varphi_k \leftarrow c\varphi_p + \varphi_k + 2c(\mathbf{x}_p \cdot \mathbf{z}_k) \\ + c(c-1)(\mathbf{z}_p \cdot \mathbf{x}_p) \end{aligned} \tag{9}$$

$$\mathbf{z}_k \leftarrow c\mathbf{z}_p + \mathbf{z}_k, \quad \mathbf{x}_k \leftarrow c\mathbf{x}_p + \mathbf{x}_k \tag{10}$$

where $\mathbf{z}_k$ in the cross term is the value *before* update.

To *scale* generator p by c (implementing $S_p \leftarrow S_p^c$), the power rule gives

$$\varphi_p \leftarrow c\varphi_p + c(c-1)(\mathbf{z}_p \cdot \mathbf{x}_p) \tag{11}$$

$$\mathbf{z}_p \leftarrow c\mathbf{z}_p, \quad \mathbf{x}_p \leftarrow c\mathbf{x}_p. \tag{12}$$

B. Unimodular Column Transformation

When we need to change basis among the generators (e.g. during measurement), we apply a unimodular matrix V to the tableau columns. Column j of the result represents the ordered product $\prod_k S_k^{V_{kj}}$. The Z and X blocks transform linearly:

$$\boldsymbol{Z}' = \boldsymbol{Z}V, \quad \boldsymbol{X}' = \boldsymbol{X}V \tag{13}$$

and the phases pick up cross terms from the ordering:

$$\begin{aligned} \varphi_j' = \sum_k V_{kj}\varphi_k + \sum_k V_{kj}(V_{kj}-1)G_{kk} \\ + 2\sum_{m<k} V_{mj} V_{kj} G_{km} \end{aligned} \tag{14}$$

where $G_{ab} = \mathbf{z}_a \cdot \mathbf{x}_b$ is the Gram matrix of the original (pre-transformation) generators.

II. Gate Conjugation

Each Clifford gate is applied by conjugating every column of the tableau: $S_j \leftarrow U S_j U^\dagger$. All phase corrections below use Z/X values *before* the update.

TABLE I: Clifford gate conjugation rules.

Gate	Z/X Update	$\Delta\varphi$
S_q	$z_q \leftarrow z_q + x_q$	Refer to (16)
F_q	$(z_q, x_q) \leftarrow (x_q, -z_q)$	$-2z_q x_q$
$F_q^\dagger$	$(z_q, x_q) \leftarrow (-x_q, z_q)$	$-2z_q x_q$
$M_{a,q}$	$z_q \leftarrow a^{-1} z_q$, $x_q \leftarrow a x_q$	0
$\mathrm{SUM}_{c\to t}$	$x_t \leftarrow x_t + x_c$, $z_c \leftarrow z_c - z_t$	0
$\mathrm{CZ}_{1,2}$	$z_1 \leftarrow z_1 + x_2$, $z_2 \leftarrow z_2 + x_1$	$2x_1 x_2$
X_q	—	$+2z_q$
Z_q	—	$-2x_q$
SWAP_{q_1,q_2}	swap rows $q_1 \leftrightarrow q_2$	0

The Fourier gate F_q swaps the Z and X roles (with a sign flip and phase correction), the multiplier gate $M_{a,q}$ rescales by $a \in \mathbb{Z}_d^\times$, and the two-qudit gates SUM and CZ act on their respective control/target pairs.

The phase gate on qudit q is defined as

$$S = \begin{cases} \sum_{q=0}^{d-1} \tau^{q^2} \ |q\rangle\langle q| & d \text{ even} \\ \sum_{q=0}^{d-1} \omega^{q(q-1)/2} \ |q\rangle\langle q| & d \text{ odd.} \end{cases} \tag{15}$$

Both forms satisfy $SZS^\dagger = Z$ (Z is unchanged) and map $X \mapsto ZX$ up to a phase absorbed into the definition. Per column:

$$z_q \leftarrow z_q + x_q, \quad \Delta\varphi = \begin{cases} x_q^2 & d \text{ even} \\ x_q(x_q+1) & d \text{ odd.} \end{cases} \tag{16}$$

III. Measurement

We want to measure a Pauli operator $P = \tau^{-\delta} Z^{\boldsymbol{a}} X^{\boldsymbol{b}}$. For a computational-basis measurement of qudit r, this is just:

$$Z_r : \text{set } \boldsymbol{a} = \hat{e}_r,\ \boldsymbol{b} = \boldsymbol{0},\ \delta = 0. \tag{17}$$

The following procedure handles both random and deterministic outcomes uniformly.

Step 1 — Commutation Vector. Compute the symplectic inner product of P against each stabilizer generator:

$$c_k = \boldsymbol{a} \cdot \mathbf{x}_k - \boldsymbol{b} \cdot \mathbf{z}_k \pmod d. \tag{18}$$

Set $\eta = \gcd(d, c_1, \ldots, c_\ell)$.

Step 2 — Isolate via SNF. Compute the Smith Normal Form of the row vector $C = [c_1 \cdots c_\ell]$ mod d, yielding a unimodular V such that

$$CV = [\eta \quad 0 \quad \cdots \quad 0]. \tag{19}$$

Apply V to the full tableau using the unimodular column transformation. After this change of basis, only column 1 has a nonzero commutation value η. Set $s = d/\eta$.

Step 3 — Eigenvalue of $\boldsymbol{P^s}$. The operator P^s commutes with the entire stabilizer group (since $s \cdot \eta \equiv 0 \bmod d$), so the stabilizer already pins down its eigenvalue. Compute P^s via the power rule:

$$\varphi_{P^s} = s\delta + s(s-1)(\boldsymbol{a} \cdot \boldsymbol{b}) \tag{20}$$

$$\mathbf{z}_{P^s} = s\boldsymbol{a}, \quad \mathbf{x}_{P^s} = s\boldsymbol{b}. \tag{21}$$

To find the eigenvalue, decompose P^s over the generators by solving $G\boldsymbol{c} \equiv \boldsymbol{v}_{P^s} \pmod d$, where G is the $2n \times \ell$ Weyl block and $\boldsymbol{v}_{P^s} = \mathbf{z}_{P^s} \oplus \mathbf{x}_{P^s}$. The SNF of G yields $UGV = D$, reducing to a diagonal system $D\boldsymbol{y} \equiv U\boldsymbol{v}_{P^s} \pmod d$ with $\boldsymbol{c} = V\boldsymbol{y}$. Record the first invariant factor $f_1 = D_{11}$.

Evaluate the ordered product $Q = \prod_k S_k^{c_k}$ using the product and power rules. The Weyl part of Q equals P^s; the phase difference $t = \varphi_Q - \varphi_{P^s} \pmod D$ encodes the eigenvalue.

Step 4 — Sample the Outcome. The measurement outcome h satisfies

$$2sh \equiv t \pmod D. \tag{22}$$

Sample h uniformly from the solution set $h_0 + \eta\mathbb{Z}_d$, which has η equally likely outcomes. When $\eta = d$, there is exactly one solution (deterministic).

Step 5 — Collapse the Tableau. Update the tableau to reflect the post-measurement state. Scale column 1 by s (via generator scaling), which zeroes out its commutation with P, and insert the measurement result $R = \tau^{-(\delta+2h)} Z^{\boldsymbol{a}} X^{\boldsymbol{b}}$ as a new generator.

If $S_1^s = \mathbb{1}$ (the scaled column is all zeros mod d), drop it. Otherwise, check whether $f_1 \mid s$. If so, S_1^s is already in the span of the other generators and can be dropped; if not, keep it (this increases ℓ by 1, up to a maximum of $2n$).

When $\eta = d$, column 1 already commutes with P, the scaling is trivial ($s = 1$), and P is already in the stabilizer group. The tableau is unchanged.

IV. References

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V. Appendix: Syndrome Extraction Circuit

Alternatively, $P = \bigotimes_j X_j^{a_j} Z_j^{b_j}$ can be measured indirectly via an ancilla r in $|0\rangle$ using the circuit

$$\Lambda P = \left(S_r^{-\boldsymbol{a}\cdot\boldsymbol{b}} \otimes \mathbf{1}\right) \prod_j \mathrm{CZ}_{r,j}^{a_j} \prod_j \mathrm{SUM}_{r\to j}^{b_j} \tag{23}$$

sandwiched by $F_r, F_r^\dagger$ on the ancilla, followed by measuring Z_r. The direct protocol above avoids this decomposition.